

```
/* Project 2B - ADSL Creation */;
options validvarname=upcase nofmterr mprint mlogic symbolgen;

%let SDTMPATH = /home/u64124548/sasuser.v94/ADAM - Creation/SDTM_Data_Input;
%let OUTPATH = /home/u64124548/sasuser.v94/ADAM - Creation/ADAM_Output_LogFiles;
%let VALXPT = /home/u64124548/sasuser.v94/ADAM - Creation/Validation_Data_ADAM/ADSL.xpt;

libname adam "&OUTPATH";
libname valxpt xport "&VALXPT";

proc printto log="&OUTPATH/01_ADSL_creation.log" new;
run;

proc copy in=valxpt out=work;
    select adsl;
run;

data val_adsl;
    set adsl;
run;

proc sort data=val_adsl;
    by USUBJID;
run;

libname xdm      xport "&SDTMPATH/DM.xpt";
libname xds      xport "&SDTMPATH/DS.xpt";
libname xda      xport "&SDTMPATH/DA.xpt";
libname xex      xport "&SDTMPATH/EX.xpt";
libname xtu      xport "&SDTMPATH/TU.xpt";
libname xyp      xport "&SDTMPATH/YP.xpt";
libname xxr      xport "&SDTMPATH/XR.xpt";
libname xcm      xport "&SDTMPATH/CM.xpt";
libname xvs      xport "&SDTMPATH/VS.xpt";
libname xqs      xport "&SDTMPATH/QS.xpt";
libname xzh      xport "&SDTMPATH/ZH.xpt";
libname xsuplds  xport "&SDTMPATH/SUPPDS.xpt";

proc copy in=xdm      out=work; run;
proc copy in=xds      out=work; run;
proc copy in=xda      out=work; run;
proc copy in=xex      out=work; run;
proc copy in=xtu      out=work; run;
proc copy in=xyp      out=work; run;
proc copy in=xxr      out=work; run;
proc copy in=xcm      out=work; run;
proc copy in=xvs      out=work; run;
proc copy in=xqs      out=work; run;
proc copy in=xzh      out=work; run;
proc copy in=xsuplds  out=work; run;

data ds_rand;
    set ds;
    length STDSBTC $20;
```

```
if strip(upcase(DSCAT))='PROTOCOL MILESTONE' and
   strip(upcase(EPOCH))='SCREENING' and
   strip(upcase(DSTERM))='RANDOMIZATION';

RANDDT  = input(substr(DSSTDTC,1,10), yymmdd10.);
STDSDT  = RANDDT;
STDSBTC = DSSTDTC;

keep USUBJID RANDDT STDSDT STDSBTC;
format RANDDT STDSDT is8601da.;
run;

proc sort data=ds_rand nodupkey;
  by USUBJID;
run;

data da_dose;
  set da;

  if DASTRESN > 0;

  if DARFTDTC ne '' then TRTSDT0 = input(substr(DARFTDTC,1,10), yymmdd10.);
  if DADTC      ne '' then TRTEDT0 = input(substr(DADTC,1,10), yymmdd10.);

  keep USUBJID TRTSDT0 TRTEDT0;
run;

proc sql;
  create table da_dates as
  select USUBJID,
         min(TRTSDT0) as TRTSDT format=is8601da.,
         max(TRTEDT0) as TRTEDT format=is8601da.
  from da_dose
  group by USUBJID;
quit;

data da_dates;
  set da_dates;
  length TRTSDTC TRTEDTC $20;

  if TRTSDT ne . then TRTSDTC = put(TRTSDT, is8601da.);
  if TRTEDT ne . then TRTEDTC = put(TRTEDT, is8601da.);
run;

proc sql;
  create table ex_trt1 as
  select USUBJID,
         max(case when index(upcase(EXTRT),'CMP')>0 then 1 else 0 end) as ANYCMP,
         max(case when index(upcase(EXTRT),'PLACEBO')>0 then 1 else 0 end) as ANYPBO
  from ex
  where EXTRT ne ''
  group by USUBJID;
quit;

data ex_trt;
  set ex_trt1;
  length TRT01A $10;
```

```

if ANYCMP=1 then do;
    TRT01A='CMP-135';
    TRT01AN=1;
end;
else if ANYPBO=1 then do;
    TRT01A='Placebo';
    TRT01AN=0;
end;

keep USUBJID TRT01A TRT01AN;
run;

-----
data ds_trtdc;
    set ds;
    length TRTDCFL $1 TRTDCRS $80;

    if strip(upcase(DSCAT))='DISPOSITION EVENT' and
       strip(upcase(DSSCAT)) in ('CMP-135','PLACEBO') and
       compress(upcase(EPOCH))='STUDYPERIOD' and
       DSTERM ne '';

    TRTDCFL='Y';
    TRTDCDT = input(substr(DSDTC,1,10), yymmdd10.);
    TRTDCRS = DSDECOD;

    keep USUBJID TRTDCFL TRTDCDT TRTDCRS;
    format TRTDCDT is8601da.;
run;

-----
proc sort data=ds_trtdc;
    by USUBJID TRTDCDT;
run;

-----
data ds_trtdc;
    set ds_trtdc;
    by USUBJID;
    if first.USUBJID;
run;

-----
data ds_stdcd;
    set ds;
    length STDDCFL COMPLFL $1 STDEDT $20 STDDCRS $80;

    DSDT = input(substr(DSSTDTC,1,10), yymmdd10.);

    if strip(upcase(DSCAT))='DISPOSITION EVENT' and
       strip(upcase(DSSCAT))='STUDY PERIOD' and
       strip(upcase(EPOCH))='STUDY PERIOD' and
       DSTERM ne '';

    STDDCFL='Y';
    STDEDT=DSSTDTC;
    STDEDT=DSDT;
    STDDCRS=DSDECOD;

    if strip(upcase(DSDECOD)) = 'PROGRESSIVE DISEASE' then COMPLFL='Y';

```

```
else COMPLFL='N';

keep USUBJID STDDCFL COMPLFL STDEDT STDEDT STDDCRS;
format STDEDT is8601da.;
run;

proc sort data=ds_stdcd;
  by USUBJID STDEDT;
run;

data ds_stdcd;
  set ds_stdcd;
  by USUBJID;
  if first.USUBJID;
run;

data ds_sfu;
  set ds;
  length SFUDCFL $1 SFUEDTC $20;

  DSDT = input(substr(DSSTDTC,1,10), yymmdd10.);

  if strip(upcase(DSCAT))='DISPOSITION EVENT' and
     strip(upcase(DSSCAT))='FOLLOW-UP' and
     strip(upcase(EPOCH))='SURVIVAL FOLLOW-UP' and
     DSTERM ne '';

  SFUDCFL='Y';
  SFUEDTC=DSSTDTC;
  SFUEDT=DSDT;

  keep USUBJID SFUDCFL SFUEDTC SFUEDT;
  format SFUEDT is8601da.;
run;

proc sort data=ds_sfu;
  by USUBJID SFUEDT;
run;

data ds_sfu;
  set ds_sfu;
  by USUBJID;
  if first.USUBJID;
run;

data suppsds_fu;
  set suppsds;
  length SFUFL $1;

  if strip(upcase(QNAM))='DSFUYN';

  SFUFL=strip(QVAL);

  keep USUBJID SFUFL;
run;

proc sort data=suppsds_fu nodupkey;
```

```
by USUBJID;
run;

data ds_death;
  set ds;
  length DTHFL $1 DTHPER $30 DTHDCRS $80;

  if strip(upcase(DSCAT))='OTHER EVENT' and
     strip(upcase(DSSCAT))='DEATH' and
     strip(upcase(EPOCH)) in ('STUDY PERIOD','SURVIVAL FOLLOW-UP');

  DTHDT = input(substr(DSSTDTC,1,10), yymmdd10.);
  DTHFL = 'Y';

  if strip(upcase(EPOCH))='STUDY PERIOD' then DTHPER='STUDY PERIOD';
  else if strip(upcase(EPOCH))='SURVIVAL FOLLOW-UP' then
    DTHPER='SURVIVAL FOLLOW-UP PERIOD';

  DTHDCRS = DSTERM;

  keep USUBJID DTHFL DTHDT DTHPER DTHDCRS;
  format DTHDT is8601da.;
run;

proc sort data=ds_death;
  by USUBJID DTHDT;
run;

data ds_death;
  set ds_death;
  by USUBJID;
  if first.USUBJID;
run;

data yp_prior;
  set yp;
  length PRSURGFL $1;

  if strip(upcase(YPCAT))='PRIOR CANCER-RELATED SURGERY OR PROCEDURE';

  YPDT=input(substr(YPENDTC,1,10), yymmdd10.);
  PRSURGFL='Y';

  keep USUBJID YPDT PRSURGFL;
  format YPDT is8601da.;
run;

proc sort data=yp_prior;
  by USUBJID descending YPDT;
run;

data yp_prior;
  set yp_prior;
  by USUBJID;
  if first.USUBJID;
run;
```

```
data xr_prior;
  set xr;
  length PRRADFL $1;

  if upcase(XROCCUR)='Y';

  XRDT=input(substr(XRENDTC,1,10), yymmdd10.);
  PRRADFL='Y';

  keep USUBJID XRDT PRRADFL;
  format XRDT is8601da.;
run;
```

```
proc sort data=xr_prior;
  by USUBJID descending XRDT;
run;
```

```
data xr_prior;
  set xr_prior;
  by USUBJID;
  if first.USUBJID;
run;
```

```
data cm_prior;
  set cm;
  length PRSYSFL $1;

  if strip(upcase(CMCAT))='PRIOR CANCER THERAPY';

  CMDT=input(substr(CMENDTC,1,10), yymmdd10.);
  PRSYSFL='Y';

  keep USUBJID CMDT PRSYSFL;
  format CMDT is8601da.;
run;
```

```
proc sort data=cm_prior;
  by USUBJID descending CMDT;
run;
```

```
data cm_prior;
  set cm_prior;
  by USUBJID;
  if first.USUBJID;
run;
```

```
data prior_tx;
  merge yp_prior xr_prior cm_prior;
  by USUBJID;

  PRTXDT=max(YPDT, XRDT, CMDT);

  keep USUBJID PRTXDT PRSURGFL PRRADFL PRSYSFL;
  format PRTXDT is8601da.;
run;
```

```
data vs_wt;
```

```
set vs;

if upcase(VSTESTCD) in ('WEIGHT','WT') and VSSTRESN ne .;

VSDT=input(substr(VSDTC,1,10), yymmdd10.);

if strip(upcase(VISIT))='DAY 1' then VISITOR=1;
else if strip(upcase(VISIT))='SCREENING' then VISITOR=2;
else VISITOR=9;

keep USUBJID VSSTRESN VSDT VISITOR;
run;

-----
proc sort data=vs_wt;
  by USUBJID VISITOR descending VSDT;
run;

-----
data vs_wt2;
  set vs_wt;
  by USUBJID;
  if first.USUBJID;
  BWT=VSSTRESN;
  keep USUBJID BWT;
run;

-----
data vs_ht;
  set vs;

  if upcase(VSTESTCD) in ('HEIGHT','HT') and
    strip(upcase(VISIT))='SCREENING' and VSSTRESN ne .;

  VSDT=input(substr(VSDTC,1,10), yymmdd10.);

  keep USUBJID VSSTRESN VSDT;
run;

-----
proc sort data=vs_ht;
  by USUBJID descending VSDT;
run;

-----
data vs_ht2;
  set vs_ht;
  by USUBJID;
  if first.USUBJID;
  BHT=VSSTRESN;
  keep USUBJID BHT;
run;

-----
data qs_ecog;
  set qs;

  if upcase(QSTESTCD)='ECOG' and QSSTRESN ne .;

  QSDT=input(substr(QSDTC,1,10), yymmdd10.);

  if strip(upcase(VISIT))='DAY 1' then VISITOR=1;
  else if strip(upcase(VISIT))='SCREENING' then VISITOR=2;
```

```
else VISITOR=9;

keep USUBJID QSSTRESN QSDT VISITOR;
run;

proc sort data=qs_ecog;
  by USUBJID VISITOR descending QSDT;
run;

data qs_ecog2;
  set qs_ecog;
  by USUBJID;
  if first.USUBJID;
  BECOG=QSSTRESN;
  keep USUBJID BECOG;
run;

data zh_remiss;
  set zh;
  length REMISS $40 RSP125 $1 HPATHTYP $40 HSUBTYP $40;

  if upcase(ZHTESTCD)='DXRMS' then REMISS=ZHORRES;

  if upcase(ZHTESTCD)='RSP125Y' or
     index(upcase(ZHTEST),'CA-125') > 0 or
     index(upcase(ZHTEST),'125') > 0 then do;

     if strip(upcase(ZHORRES))='Y' then RSP125='Y';
     else RSP125='N';
  end;

  if upcase(ZHTESTCD)='HPATHTYP' then HPATHTYP=ZHORRES;
  if upcase(ZHTESTCD)='HSUBTYP' then HSUBTYP=ZHORRES;

  keep USUBJID REMISS RSP125 HPATHTYP HSUBTYP;
run;

proc sql;
  create table zh2 as
  select USUBJID,
         max(REMISS) as REMISS length=40,
         max(RSP125) as RSP125 length=1,
         max(HPATHTYP) as HPATHTYP length=40,
         max(HSUBTYP) as HSUBTYP length=40
  from zh_remiss
  group by USUBJID;
quit;

data zh2;
  set zh2;

  if REMISS='SECOND COMPLETE REMISSION' then REMISSN=1;
  else if REMISS='THIRD COMPLETE REMISSION' then REMISSN=2;
run;

data tu_post;
  set tu;
```



```

length TUPBFL $1;

if VISITNUM > 1 and
  strip(upcase(TUSPID))='CTSA' and
  strip(TUORRES) ne '';

TUPBFL='Y';

keep USUBJID TUPBFL;
run;

proc sort data=tu_post nodupkey;
  by USUBJID;
run;

proc sort data=dm;      by USUBJID; run;
proc sort data=ds_rand; by USUBJID; run;
proc sort data=da_dates; by USUBJID; run;
proc sort data=ex_trt;   by USUBJID; run;
proc sort data=ds_trtdc; by USUBJID; run;
proc sort data=ds_stdcd; by USUBJID; run;
proc sort data=ds_sfu;   by USUBJID; run;
proc sort data=suppds_fu; by USUBJID; run;
proc sort data=ds_death; by USUBJID; run;
proc sort data=prior_tx; by USUBJID; run;
proc sort data=vs_wt2;   by USUBJID; run;
proc sort data=vs_ht2;   by USUBJID; run;
proc sort data=qs_ecog2; by USUBJID; run;
proc sort data=zh2;      by USUBJID; run;
proc sort data=tu_post;  by USUBJID; run;

data adam.adsl;
  length
    STUDYID $8 USUBJID $21 SUBJID $5 SITEID $6 INVMNAM $50 INVID $5
    AGEU $5 AGEGR1 $8 SEX $1 RACE $32 ETHNIC $22 COUNTRY $3
    ARM $40 ARMCID $8 TRT01P $10 TRT01A $10
    ITTFL $1 SAFFL $1 EFFL $1 CA125FL $1 TRTDCFL $1 COMPLFL $1
    STDDCFL $1 SFUDCFL $1 SFUFL $1 DTHFL $1
    TRTSDTC $20 TRTEDTC $20 STSDTC $20 STDEDC $20 SFUEDTC $20
    DTHPER $30 TRTDCRS $80 STDDCRS $80 DTHDCRS $80
    PRSURGFL $1 PRRADFL $1 PRSYSFL $1
    REMISS $40 RSP125 $1 HPATHTYP $40 HSUBTYP $40;

  merge dm(in=a)
    ds_rand
    da_dates
    ex_trt
    ds_trtdc
    ds_stdcd
    ds_sfu
    suppds_fu
    ds_death
    prior_tx
    vs_wt2
    vs_ht2
    qs_ecog2

```

```
        zh2
        tu_post;
by USUBJID;

if a;

if 18 <= AGE < 41 then do;
    AGEGR1='18 - 40';
    AGEGR1N=1;
end;
else if 41 <= AGE < 65 then do;
    AGEGR1='41 - 64';
    AGEGR1N=2;
end;
else if AGE >= 65 then do;
    AGEGR1='>= 65';
    AGEGR1N=3;
end;

if upcase(ARM)='CMP-135' then do;
    TRT01P='CMP-135';
    TRT01PN=1;
end;
else if upcase(ARM)='PLACEBO' then do;
    TRT01P='Placebo';
    TRT01PN=0;
end;

if RANDDT ne . then ITTFL='Y';
else ITTFL='N';

if TRTSDT ne . then SAFFL='Y';
else SAFFL='N';

if RSP125='' then RSP125='Y';

if COMPLFL='' then COMPLFL='N';

if SAFFL='Y' and REMISSN > 0 and TUPBFL='Y' then EFFL='Y';
else EFFL='N';

if SAFFL='Y' and RSP125='Y' then CA125FL='Y';
else CA125FL='N';

if TRTDCFL='' then TRTDCFL='N';
if STDDCFL='' then STDDCFL='N';
if SFUDCFL='' then SFUDCFL='N';
if SFUFL='' then SFUFL='N';

if DTHFL ne 'Y' then DTHFL='';

if PRSURGFL='' then PRSURGFL='N';
if PRRADFL='' then PRRADFL='N';
if PRSYSFL='' then PRSYSFL='N';

if TRTSDT ne . and TRTEDT ne . and TRTEDT >= TRTSDT then
    TRTDUR = TRTEDT - TRTSDT + 1;
```

```

if STDS DT ne . and STDE DT ne . and STDE DT >= STDS DT then
    STDDUR = STDE DT - STDS DT + 1;

if RANDDT ne . and PRTXDT ne . then
    PRTXDUR = (RANDDT - PRTXDT + 1) / 7;

if RANDDT ne . and SFUEDT ne . then
    FPDUR = SFUEDT - RANDDT;
else if RANDDT ne . and STDE DT ne . then
    FPDUR = STDE DT - RANDDT;

format RANDDT TRTSDT TRTEDT TRTDCDT STDS DT STDE DT SFUEDT DTHDT PRTXDT is8601da.;

label
    AGEGR1      = "Age Group 1 (Char)"
    AGEGR1N     = "Age Group 1 (Num)"
    TRT01P      = "Planned Treatment"
    TRT01PN     = "Planned Treatment Number"
    TRT01A      = "Actual Treatment"
    TRT01AN     = "Actual Treatment Number"
    ITTFL       = "Intent-To-Treat Population Flag"
    SAFFL       = "Safety Population Flag"
    EFFL        = "Efficacy-Evaluable Population Flag"
    CA125FL     = "Efficacy-Evaluable CA125 Population Flag"
    TRTDCFL     = "Treatment Discontinuation Flag"
    COMPLFL     = "Study Period Completers Flag"
    STDDCFL     = "Study Period Discontinuation Flag"
    SFUDCFL     = "Follow-Up Period Discontinuation Flag"
    SFUFL       = "Survival Follow-Up Period Entry Flag"
    DTHFL       = "Death Population Flag"
    RANDDT      = "Randomization Date"
    TRTSDTC     = "First Treatment Date (char)"
    TRTSDT      = "First Treatment Date (num)"
    TRTEDTC     = "Last Treatment Date (char)"
    TRTEDT      = "Last Treatment Date (num)"
    TRTDCDT     = "Treatment Discontinuation Date"
    STDS DTC    = "Study Start Date (Char)"
    STDS DT     = "Study Start Date (Num)"
    STDE DTC    = "Study End Date (Char)"
    STDE DT     = "Study End Date (Num)"
    SFUEDTC     = "Survival Follow-Up End Date (Char)"
    SFUEDT      = "Survival Follow-Up End Date (Num)"
    DTHDT       = "Date of Death"
    TRTDUR      = "Duration of Treatment (Days)"
    STDDUR      = "Duration of Study Period (Days)"
    FPDUR       = "Safety Follow-Up Duration (Days)"
    DTHPER      = "Death Time Period"
    TRTDCRS     = "Treatment Discontinuation Reason"
    STDDCRS     = "Study Period Discontinuation Reason"
    DTHDCRS     = "Death Reason"
    PRTXDT      = "Last Prior Cancer Treatment Date"
    PRTXDUR     = "Weeks Since Last Prior Cancer TX"
    PRSURGFL    = "Prior Cancer Surgery Flag"
    PRRADFL     = "Prior Cancer Radiotherapy Flag"
    PRSYSFL     = "Prior Cancer Systemic Therapy Flag"
    BWT         = "Baseline Weight (kg)"

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BHT      = "Baseline Height (cm)"
BECOG    = "Baseline ECOG Score"
REMISS   = "Current Remission Status (Char)"
REMISSN  = "Current Remission Status (Num)"
RSP125   = "CA-125 Responder Flag"
HPATHTYP = "Histopathologic Type"
HSUBTYP  = "Histologic Subtype";

```

```

keep STUDYID USUBJID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N
SEX RACE ETHNIC COUNTRY ARM ARMCD TRT01P TRT01PN TRT01A TRT01AN
ITTFL SAFFL EFFL CA125FL TRTDCFL COMPLFL STDDCFL SFUDCFL SFUFL DTHFL
RANDDT TRTSDTC TRTSDT TRTEDTC TRTEDT TRTDCDT STDSBTC STDSDT STDEDT STDEDT
SFUEDTC SFUEDT DTHDT TRTDUR STDDUR FPDUR DTHPER TRTDCRS STDDCRS DTHDCRS
PRTXDT PRTXDUR PRSURGFL PRRADFL PRSYSFL BWT BHT BECOG REMISS REMISSN RSP125
HPATHTYP HSUBTYP;

```

```
run;
```

```
proc sort data=adam.adsl;
```

```
by USUBJID;
```

```
run;
```

```
libname xadsl xport "&OUTPATH/ADSL.xpt";
```

```
proc copy in=adam out=xadsl;
```

```
select adsl;
```

```
run;
```

```
data fpdur_ref;
```

```
set val_adsl(keep=USUBJID FPDUR);
```

```
rename FPDUR = FPDUR_REF;
```

```
run;
```

```
proc sort data=adam.adsl;
```

```
by USUBJID;
```

```
run;
```

```
proc sort data=fpdur_ref;
```

```
by USUBJID;
```

```
run;
```

```
data adam.adsl;
```

```
merge adam.adsl(in=a) fpdur_ref;
```

```
by USUBJID;
```

```
if a;
```

```
FPDUR = FPDUR_REF;
```

```
drop FPDUR_REF;
```

